



NALAM NOW :)





ZENITH

PROBLEM STATEMENT



Municipal Waste Mismanagement in Urban Areas

- Overflowing public garbage bins
- Waste thrown outside bins
- Lack of real-time monitoring
- Manual inspection by municipality staff
- No strict enforcement against littering
- Citizens unaware of proper disposal practices
- Poor complaint redressal system



Impact:

- Unhygienic streets
- Spread of diseases
- Bad public image
- Increased municipal workload



EXISTING SOLUTIONS & THEIR DRAWBACKS

01

Traditional Public Bins

- Simple bins placed in public locations
- ✗ No monitoring
- ✗ No alert when full
- ✗ No misuse detection

02

Manual Inspection System

- Workers check bins periodically
- ✗ Time-consuming
- ✗ Delayed response
- ✗ Inefficient resource allocation

03

Basic Complaint Systems

- Citizens report through calls or apps
- ✗ Slow action
- ✗ No live tracking
- ✗ Poor transparency



PROPOSED SOLUTION

Smart Waste Governance Ecosystem

An integrated, technology-driven waste management system combining:
Smart IoT Garbage Bins

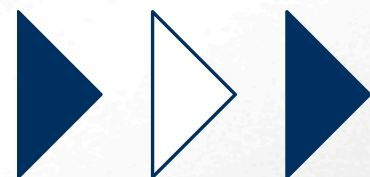
- Sensors detect:
 - Waste thrown outside bin
 - Bin fill level
- Tamil voice warning system
- Red alert light for improper disposal
- Data logging of misuse

Municipal Monitoring Dashboard

- Real-time bin status
- Alerts for overflow
- Location-based tracking
- Data analytics

Citizen Complaint PWA App

- Upload photo of litter
- Geo-location tagging
- Live complaint status tracking





METHODOLOGY

Step 1: Smart Bin Development

- Install ultrasonic sensors for fill-level detection
- Use IR sensors for outside-waste detection
- Microcontroller (ESP32/Arduino) for processing
- GSM/WiFi module for data transmission

Step 2: Backend System

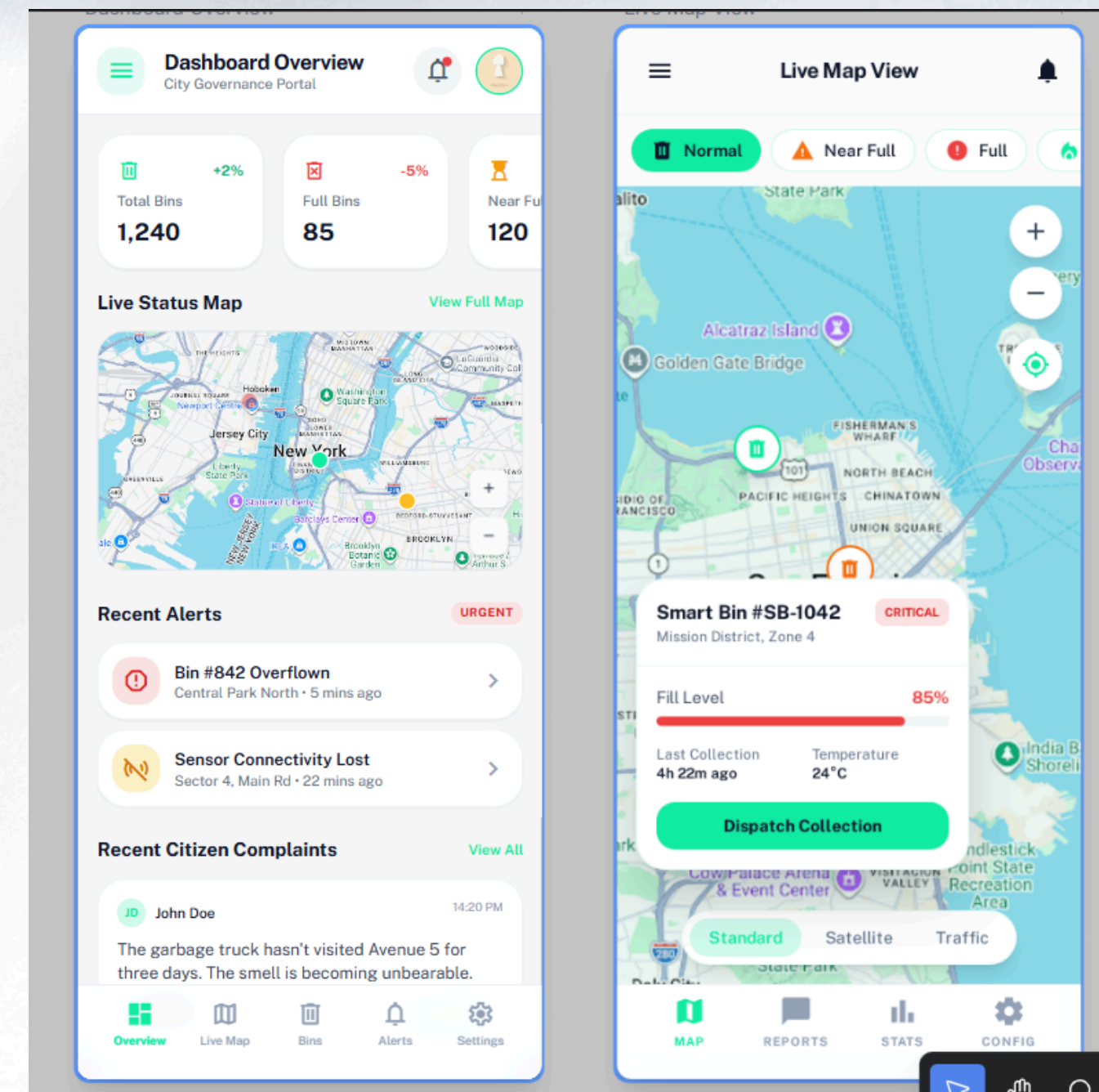
- Cloud database for storing bin data
- API development
- Real-time dashboard for municipality

Step 3: PWA Mobile Application

- Complaint registration
- Image upload
- GPS tagging
- Status tracking

Step 4: Data Analytics

- Identify high-littering areas
- Predict cleaning schedules
- Optimize workforce deployment





ZENITH

ESTIMATED COST

Smart Components Added to Existing Bins		
Component	Purpose	Approx Cost (INR)
Ultrasonic Sensor	Detect bin fill level	₹250
IR Sensor	Detect waste thrown outside	₹150
ESP32 Microcontroller	Process & send data	₹600
GSM/WiFi Module	Internet connectivity	₹700
Speaker + Voice Module	Tamil warning alert	₹500
LED Alert Light	Visual misuse alert	₹200
Power Supply & Battery	System power	₹800
Wiring & Installation	Fitting & setup	₹500
💰 Total Estimated Cost Per Bin (Integration Only)		
👉 ₹3,700 – ₹4,000 (Approx)		

85%
SALES



THANK YOU

